

Zekun Song

Backend & Data Engineer | Clinical AI | Reliable Data Platforms

GPA: 3.87/4.00 | 8-month co-op: Sep 2026 - Apr 2027 | Canadian permanent resident - no sponsorship required

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EDUCATION

University of Toronto (St. George)
Honours Bachelor of Science (HBSoc) - Double Specialist in Computer Science and Data Science
Honours: Dean's List Scholar; Reuben Wells Leonard Scholarship
Coursework: Data Structures & Algorithms, Systems Programming in C and Unix/Shell, Databases and SQL, Software Engineering, Machine Learning, Data Science I & II, Time Series Analysis

Sep 2023 - Dec 2027 (Expected)
Toronto, ON

EXPERIENCE

Remeda - Hospital-Partnered Clinical AI Startup
Backend & Data Engineer | Clinical AI Platform (CSC494 Venture)

Apr 2026 - Present
Toronto, ON

- Clinical Knowledge Layer:** Own the terminology and reference-data layer of an OMOP CDM PostgreSQL warehouse powering a clinical AI platform, integrating OpenMRS records via FHIR with NLP, Synthea, and MIMIC-IV data.
- Terminology ETL at Scale:** Integrated 12 controlled vocabularies and 90,000+ clinical codes via APIs, bulk loads, and Python ETL; built a FastAPI/asyncpg service mapping free-text entities to SNOMED CT, ICD-10, RxNorm, and LOINC.
- Data Reliability:** Made ingestion rerun-safe with idempotent migrations, batched SQL writes, SHA-256-pinned reference data, and 60+ unit tests; blocked an underdetermined metric from shipping after checking it against CMS specifications.
- Production Debugging:** Root-caused a demo-blocking stale backend address before the CanHealth industry demo, then deployed, verified end to end, and co-presented the platform.

Shell (Shanghai) Technology Co., Ltd.
Data Science Intern

Jun 2024 - Sep 2024
Shanghai, China

- Self-Service Analytics:** Delivered an internal portal whose chat interface turns business questions into governed SQL, cutting weekly ad-hoc data requests from about 16 to 7.
- NL-to-SQL Validation:** Paired LLM query generation with schema-grounding and permission checks against a production database; 92% of 300+ benchmark queries produced SQL that satisfied the validation policy.
- Instrumentation & BI:** Built query-logging and user-analytics modules plus 10+ auto-refreshed Power BI dashboards over SQL pipelines, cutting the product-feedback cycle from three weeks to one.

WholeViz - Industry Partner Capstone
Technical Product Lead | Full-Stack Analytics Platform

Sep 2025 - Dec 2025
Toronto, ON

- Team Leadership:** Led a 6-person team shipping a Django, TypeScript, and PostgreSQL analytics platform for non-technical users, owning backend architecture and delivery through industry-partner handoff.
- Backend & Data:** Designed normalized schemas, multi-user access management, and automated ingestion, validation, and storage pipelines under pytest; translated stakeholder needs into specifications through agile iterations.

PROJECTS

CardioTrace | *Python, PostgreSQL, dbt, Survival Analysis*

Jun 2026 - Present

- Cohort Engineering & Modelling:** Built a reproducible NHANES-mortality pipeline spanning 11 survey cycles and 235,553 person-years; trained survey-weighted competing-risk Cox models on 20,736 adults and 925 cardiovascular deaths using a forward-time split, reaching a 10-year C-index of 0.804.
- Data Integrity:** Found CDC renames dropping lab data for 15,332 people; withdrew results and rebuilt 64-test ingestion.

Personal Job Application OS | *FastAPI, Next.js, SQLAlchemy*

Jul 2026 - Present

- Platform Engineering:** Built a FastAPI and Next.js platform that normalizes 10,921 postings from 58 first-party ATS sources behind a 63-table SQLAlchemy/Alembic schema with an append-only audit trail.
- Decision Engine & Safety:** Implemented deduplication, eligibility gates, role classification, and evidence-weighted ranking; enforced hash-bound per-application confirmation, with 280 pytest cases passing.

TECHNICAL SKILLS

Languages Python, SQL, TypeScript/JavaScript, R, Java, Bash	Backend & Cloud FastAPI, Django, REST APIs, asyncio/asyncpg, SQLAlchemy/Alembic, Docker, Vercel, Render, Supabase	Data & AI PostgreSQL, dbt, Pandas, NumPy, Scikit-learn, XGBoost, data modelling, ETL pipelines, statistical modelling, LLM/NL-to-SQL evaluation	Delivery & BI Git/GitHub, pytest, GitHub Actions (CI/CD), Linux, Power BI, Tableau, Plotly, Excel
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